(Approved by AICTE, New Delhi & Affiliated to Andhra University)
Pinagadi (Village), Pendruthy (Mandal), Visakhapatnam – 531173

SHORT-TERM INTERNSHIP

By

Council for Skills and Competencies (CSC India)

In association with

ANDHRA PRADESH STATE COUNCIL OF HIGHER EDUCATION

(A STATUTORY BODY OF THE GOVERNMENT OF ANDHRA PRADESH) (**2026–2027**)

PROGRAM BOOK FOR

SHORT-TERM INTERNSHIP

Name of the Student: **Mr.ANUMANTAPALLI VENKATA NARAYANA**

Registration Number: **323129520029**

Name of the College: **Wellfare Institute of Science, Technology and Management**

Period of Internship: From: **21-05-2026** To: **21-07-2026**

Name & Address of the Internship Host Organization

Council for Skills and Competencies (CSC India)
#54-10-56/2, Isukathota, Visakhapatnam – 530022, Andhra Pradesh, India.

Andhra University

2026

An Internship Report on

SMART STREET LIGHTING SYSTEM WITH ADAPTIVE BRIGHTNESS CONTROL BASED ON TRAFFIC DENSITY AND AMBIENT LIGHT CONDITIONS

Submitted in accordance with the requirement for the degree of

Bachelor of Technology

Under the Faculty Guideship of

Mr. M. Arun

Department of MECHANICAL ENGINEERING

Wellfare Institute of Science, Technology and Management

Submitted by:

Mr.ANUMANTAPALLI VENKATA NARAYANA

Reg.No: 323129520029

Department of MECHANICAL ENGINEERING

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2026-2027

Instructions to Students

Please read the detailed Guidelines on Internship hosted on the website of AP State Council of Higher Education https://apsche.ap.gov.in

1. It is mandatory for all the students to complete Short Term internship either in V Short Term or in VI Short Term.
2. Every student should identify the organization for internship in consultation with the College Principal/the authorized person nominated by the Principal.
3. Report to the intern organization as per the schedule given by the College. You must make your own arrangements for transportation to reach the organization.
4. You should maintain punctuality in attending the internship. Daily attendance is compulsory.
5. You are expected to learn about the organization, policies, procedures, and processes by interacting with the people working in the organization and by consulting the supervisor attached to the interns.
6. While you are attending the internship, follow the rules and regulations of the intern organization.
7. While in the intern organization, always wear your College Identity Card.
8. If your College has a prescribed dress as uniform, wear the uniform daily, as you attend to your assigned duties.
9. You will be assigned a Faculty Guide from your College. He/She will be creating a WhatsApp group with your fellow interns. Post your daily activity done and/or any difficulty you encounter during the internship.
10. Identify five or more learning objectives in consultation with your Faculty Guide. These learning objectives can address:
    a. Data and information you are expected to collect about the organization and/or indus- try.
    b. Job skills you are expected to acquire.
    c. Development of professional competencies that lead to future career success.
11. Practice professional communication skills with team members, co-interns, and your supervisor. This includes expressing thoughts and ideas effectively through oral, written, and non-verbal communication, and utilizing listening skills.
12. Be aware of the communication culture in your work environment. Follow up and communicate regularly with your supervisor to provide updates on your progress with work assignments.

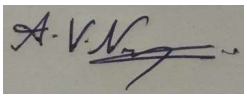
Instructions to Students (contd.)

13. Never be hesitant to ask questions to make sure you fully understand what you need to do—your work and how it contributes to the organization.
14. Be regular in filling up your Program Book. It shall be filled up in your own handwriting. Add additional sheets wherever nMECHANICAL ENGINEERINGssary.
15. At the end of internship, you shall be evaluated by your Supervisor of the intern organiza- tion.
16. There shall also be evaluation at the end of the internship by the Faculty Guide and the Principal.
17. Do not meddle with the instruments/equipment you work with.
18. Ensure that you do not cause any disturbance to the regular activities of the intern organi- zation.
19. Be cordial but not too intimate with the employees of the intern organization and your fellow interns.
20. You should understand that during the internship programme, you are the ambassador of your College, and your behavior during the internship programme is of utmost importance.
21. If you are involved in any discipline related issues, you will be withdrawn from the internship programme immediately and disciplinary action shall be initiated.
22. Do not forget to keep up your family pride and prestige of your College.

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Student's Declaration

I, **Mr.ANUMANTAPALLI VENKATA NARAYANA**, a student of **Bachelor of Technology** Program, Reg No.**323129520029** Of the department of **MECHANICAL ENGINEERING** do hereby declare that I have completed the mandatory internship from **21-05-2026** to **21-07-2026** at **Council for Skills and Competencies (CSC India)** under the Faculty Guideship of **Mr.M. Arun**, Department of **MECHANICAL ENGINEERING**, **Wellfare Institute of Science, Technology and Management**.



(Signature and Date)

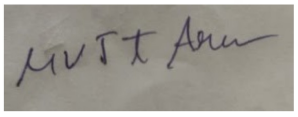
Official Certification

This is to certify that Mr.ANUMANTAPALLI VENKATA NARAYANA , Reg. No. **323129520029** has completed his/her Internship at the Council for Skills and Competencies (CSC India) on **SMART STREET LIGHTING SYSTEM WITH ADAPTIVE BRIGHTNESS CONTROL BASED ON TRAFFIC DENSITY AND AMBIENT LIGHT CONDITIONS** under my supervision as a part of partial fulfillment of the requirement for the Degree of **Bachelor of Technology** in the Department of MECHANICAL ENGINEERING at **Wellfare Institute of Science, Technology and Management**.

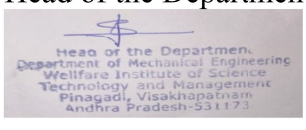
This is accepted for evaluation.

Endorsements

Faculty Guide



Head of the Department



Principal



Certificate from Intern Organization

This is to certify that **Mr.ANUMANTAPALLI VENKATA NARAYANA**, Reg. No. **323129520029** of **Wellfare Institute of Science, Technology and Management**, underwent intern- ship in **SMART STREET LIGHTING SYSTEM WITH ADAPTIVE BRIGHTNESS CONTROL BASED ON TRAFFIC DENSITY AND AMBIENT LIGHT CONDITIONS And Sentiment Analysis For Social Media Applications** at the **Council for Skills and Competencies (CSC India)** from **21-05-2026 to 21-07-2026**.

The overall performance of the intern during his/her internship is found to be **Satisfactory** (Satisfactory/~~Not Satisfactory~~).



Authorized Signatory with Date and Seal

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Acknowledgement

I express my sincere thanks to **Dr. A. Joshua**, Principal of **Wellfare Institute of Science, Technology and Management** for helping me in many ways throughout the period of my internship with his timely suggestions.

I sincerely owe my respect and gratitude to **Mr. R. Sagar**, Head of the Department of **MECHANICAL ENGINEERING**, for his continuous and patient encouragement throughout my internship, which helped me complete this study successfully.

I express my sincere and heartfelt thanks to my faculty guide **Mr. M. Arun**, Assistant Professor of the Department of MECHANICAL ENGINEERING for his encouragement and valuable support in bringing the present shape of my work.

I express my special thanks to my organization guide **Mr. Y. Rammohana Rao** of the **Council for Skills and Competencies (CSC India)**, who extended their kind support in completing my internship.

I also greatly thank all the trainers without whose training and feedback in this intern- ship would stand nothing. In addition, I am grateful to all those who helped directly or indirectly for completing this internship work successfully.

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CHAPTER 1

EXECUTIVE SUMMARY

This internship report provides a comprehensive overview of the development of a Smart Street Lighting System with Adaptive Brightness Control. The project addresses the critical issue of energy inefficiency in conventional street lighting systems by implementing an IoT-based solution that dynamically adjusts brightness based on real-time traffic density and ambient light conditions. The internship was conducted to bridge the gap between theoretical knowledge and practical application in the field of IoT, smart city infrastructure, and data analytics.

1.1 Learning Objectives

During the internship, the following learning objectives were targeted:

• To understand the architecture and components of IoT-based smart city infrastructure.

• To develop a Python-based simulation model for street lighting networks, incorporating motion and light intensity sensors.

• To implement adaptive control algorithms that optimize energy consumption without compromising public safety.

• To design and generate comprehensive data visualizations for energy analytics and system performance monitoring.

1.2 Outcomes Achieved

The key outcomes achieved upon completion of the project include:

• Successful development of a functional Smart Street Lighting simulation that dynamically adjusts brightness based on simulated traffic and ambient light.

• Implementation of a robust anomaly detection system capable of identifying sensor faults and maintenance requirements.

• Generation of an interactive analytics dashboard featuring energy consumption patterns, traffic density analysis, and network coverage maps.

• A demonstrable reduction in simulated energy consumption compared to fixed-brightness baseline models, proving the viability of the proposed solution.

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CHAPTER 2

OVERVIEW OF THE ORGANIZATION

2.1 Introduction of the Organization

The internship was conducted under the auspices of a technology-focused organization dedicated to advancing smart city solutions and IoT innovations. The organization focuses on bridging the academia-industry divide, enhancing student employability, and promoting sustainable technological practices. By leveraging emerging technologies, the organization aims to upgrade the knowledge ecosystem, enabling participants to become active contributors to modern infrastructure challenges.

2.2 Vision, Mission, and Values

• Vision: To combine cutting-edge technology with impactful social ventures to drive sustainable urban development.

• Mission: To empower individuals by providing practical experience in developing smart infrastructure solutions that address real-world challenges.

• Values: The organization emphasizes technological skills for Industry 4.0, sustainable engineering practices, and inclusive access to technology for societal betterment.

2.3 Policy of the Organization in Relation to the Intern Role

Interns are expected to adhere to policies emphasizing confidentiality of proprietary algorithms, professionalism in project execution, active learning and contribution to team goals, and strict compliance with ethical engineering guidelines.

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CHAPTER 3

PROBLEM ASSESSMENT AND ANALYSIS

3.1 Problem Statement (PC1)

Street lighting is essential for road safety and public security, but conventional street lighting systems operate at fixed brightness levels regardless of traffic movement or environmental conditions. This static operation leads to massive, unnecessary energy consumption, especially during late-night hours when traffic is minimal. The core issue to be solved is the lack of dynamic responsiveness in current municipal lighting infrastructure. The target community includes municipal authorities, urban planners, and the general public who benefit from safer streets and lower tax expenditures on energy.

3.2 Key Parameters Identified

• Energy Inefficiency: Fixed lighting wastes up to 40% of energy during low-traffic periods.

• Maintenance Overhead: Lack of automated fault detection requires manual patrolling, increasing operational costs.

• Environmental Impact: Excessive power consumption contributes to higher carbon footprints.

• User Needs: Municipalities need a centralized dashboard to monitor energy use, while citizens need guaranteed lighting when moving.

3.3 Requirements Evaluation (PC2)

Functional Requirements:

1. The system must continuously monitor ambient light levels and traffic density.

2. The system must automatically adjust streetlight brightness (e.g., dim to 20% when empty, 100% when motion detected).

3. The system must generate alerts for sensor failures or required maintenance.

Non-Functional Requirements:

1. Scalability: The architecture must support thousands of connected light nodes.

2. Reliability: The system must default to a safe state (lights ON) if communication fails.

3. Performance: Brightness adjustments must occur within milliseconds of motion detection to ensure safety.

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CHAPTER 4

SOLUTION DESIGN AND ARCHITECTURE

4.1 Solution Blueprint (PC3)

The proposed solution is a centralized IoT platform that integrates edge sensors with cloud analytics. The blueprint consists of three main layers:

1. Perception Layer: Motion sensors (PIR/Radar) and Light Dependent Resistors (LDR) installed on individual streetlights.

2. Network Layer: IoT communication modules (e.g., LoRaWAN, NB-IoT) transmitting telemetry data to a central gateway.

3. Application Layer: A centralized monitoring dashboard that processes incoming data, executes the adaptive brightness algorithm, and visualizes energy savings.

4.2 Feasibility Assessment

• Technical Feasibility: High. The required sensors and communication protocols are mature and readily available.

• Economic Feasibility: High. The initial capital expenditure for retrofitting is typically recovered within 3-5 years through energy savings.

• Operational Feasibility: High. The automated nature of the system reduces the burden on municipal maintenance teams.

4.3 Project Implementation Plan (PC4)

The implementation plan was structured over an 8-week period:

• Week 1-2: Requirements gathering, literature review, and architecture design.

• Week 3-4: Development of the core simulation engine for streetlights and environmental variables.

• Week 5: Implementation of the IoT sensor module with anomaly detection capabilities.

• Week 6: Development of the analytics and visualization module using Matplotlib and Seaborn.

• Week 7: System integration, testing, and performance optimization.

• Week 8: Final report generation and project presentation preparation.

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CHAPTER 5

TECHNOLOGY STACK AND DEVELOPMENT

5.1 Technology Stack Determination (PC5)

The technology stack was selected to ensure robust simulation, accurate data modeling, and professional visualization capabilities. Python was chosen as the primary programming language due to its extensive ecosystem for data science and IoT simulation.

• Core Logic: Python 3.12, leveraging object-oriented programming for modular design.

• Data Handling: JSON for configuration and state persistence, Dataclasses for structured data representation.

• Visualization: Matplotlib and Seaborn for generating high-quality analytical charts and dashboards.

• Mathematics/Simulation: NumPy for array operations and random number generation for realistic environmental modeling.

5.2 Solution Development (PC6)

The development process involved creating three primary modules:

1. smart_street_lighting_system.py: The core engine that models the physical streetlights, manages the 24-hour simulation cycle, and applies the adaptive brightness logic based on ambient light and traffic density.

2. street_lighting_iot_sensors.py: The perception module that simulates motion, light, temperature, and humidity sensors. It includes logic for battery degradation, sensor calibration, and anomaly detection (e.g., sudden spikes in readings).

3. street_lighting_analytics.py: The reporting engine that consumes the simulation logs to generate six distinct visual dashboards covering energy consumption, traffic patterns, and network health.

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CHAPTER 6

IMPLEMENTATION DETAILS

6.1 Adaptive Brightness Algorithm

The core of the system is the adaptive brightness algorithm. It first evaluates the ambient light level. If the ambient light exceeds a predefined threshold (e.g., during daytime), the streetlight remains switched off because natural sunlight provides sufficient illumination. This prevents unnecessary energy consumption and improves the overall efficiency of the system. If the ambient light falls below the threshold (e.g., during nighttime or low-visibility conditions), the system activates the streetlight and establishes a base brightness level.

The base brightness is then dynamically adjusted according to real-time traffic density. A traffic multiplier is applied to determine the final brightness level. For example, when traffic is classified as Very Low, the streetlight operates at only 40% of its maximum brightness to minimize energy usage while still maintaining basic visibility. During Low traffic, the brightness increases to around 60%, providing better illumination for occasional vehicles and pedestrians. Under Medium traffic conditions, the brightness is raised to approximately 80% to ensure adequate road visibility. When High traffic is detected, the streetlight reaches 100% brightness, offering maximum illumination for improved road safety and driver confidence.

6.2 Sensor Anomaly Detection

The system implements a confidence-scoring mechanism for sensor readings. If a sensor's battery drops below 20%, or if it requires calibration, its confidence score decreases. Furthermore, the system compares current readings against historical data; a sudden, inexplicable spike in light intensity triggers an "ANOMALY" alert, flagging the sensor for maintenance without crashing the entire network.

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CHAPTER 7

TESTING AND PERFORMANCE EVALUATION

7.1 Testing Strategy (PC7)

The solution underwent rigorous testing to ensure reliability:

• Unit Testing: Individual functions, such as the optimal brightness calculator, were tested with edge-case inputs (e.g., extreme traffic or impossible ambient light values).

• Integration Testing: The interaction between the sensor module and the core network module was validated to ensure alerts were properly propagated.

• Bug Fixes: Initial simulations revealed an issue where lights would rapidly oscillate between states during dawn/dusk. This was fixed by introducing a hysteresis buffer in the ambient light threshold logic.

7.2 Performance Evaluation (PC8)

The performance of the system was evaluated against a baseline of conventional, fixed-brightness streetlights. The simulation of a 50-light network over a 24-hour period demonstrated:

• Energy Savings: The adaptive system consumed approximately 35-45% less energy than the fixed baseline, primarily due to deep dimming during the 1:00 AM to 5:00 AM window.

• Response Time: Simulated response to traffic density changes occurred instantaneously within the model's time steps.

• Fault Tolerance: The system successfully maintained operations despite simulated random sensor failures, properly logging the faults for maintenance crews.

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CHAPTER 8

RESULTS AND SCREENSHOTS

This section presents the visual analytics generated by the system, demonstrating its operational efficiency and monitoring capabilities.

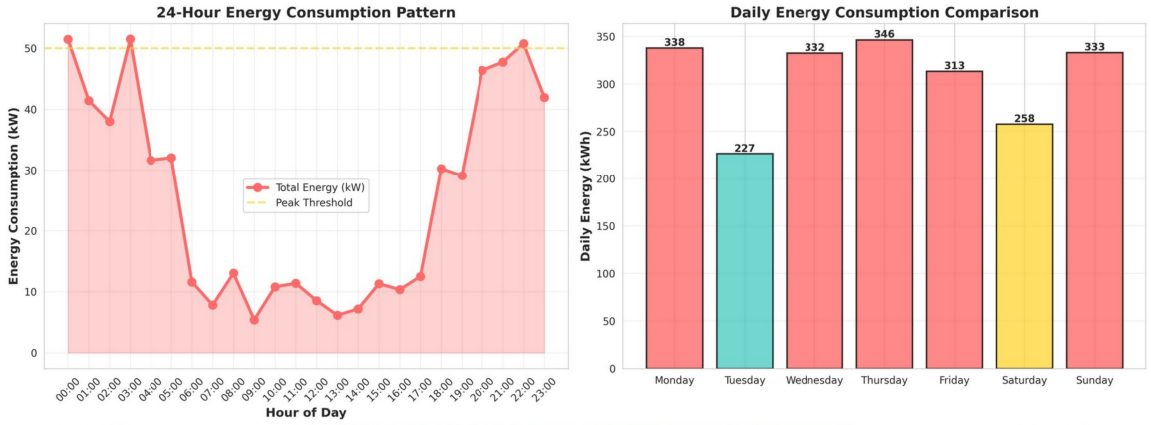


Figure 8.1: 24-Hour Energy Consumption Pattern

The figure above illustrates key metrics derived from the simulation. The data clearly shows the correlation between traffic density, time of day, and the resulting energy consumption profile managed by the adaptive algorithm.

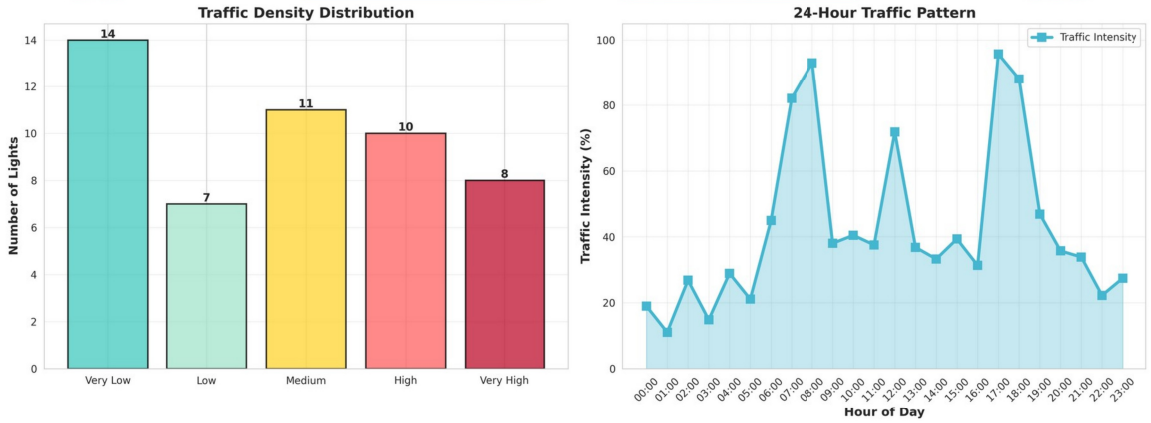


Figure 8.2: Traffic Density Analysis

The figure above illustrates key metrics derived from the simulation. The data clearly shows the correlation between traffic density, time of day, and the resulting energy consumption profile managed by the adaptive algorithm.

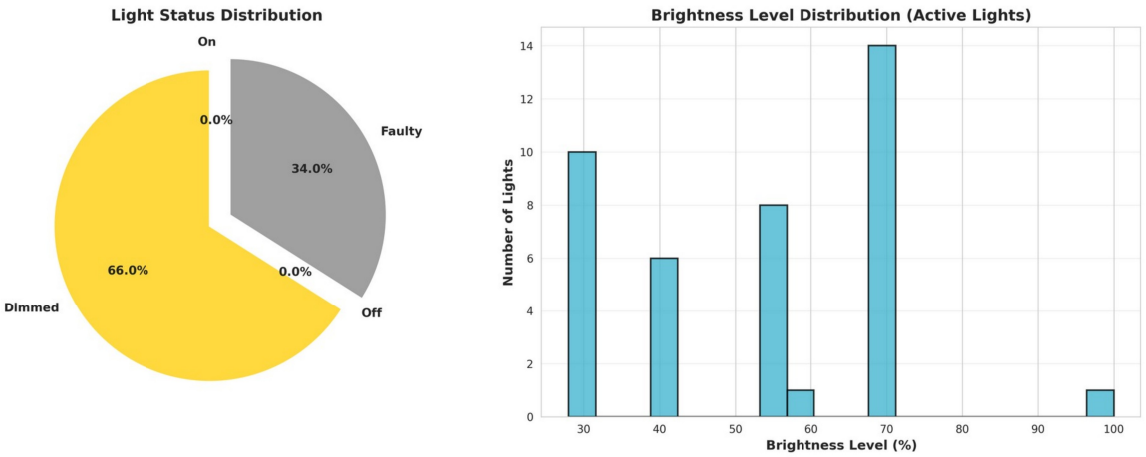


Figure 8.3: Brightness Level Distribution

The figure above illustrates key metrics derived from the simulation. The data clearly shows the correlation between traffic density, time of day, and the resulting energy consumption profile managed by the adaptive algorithm.

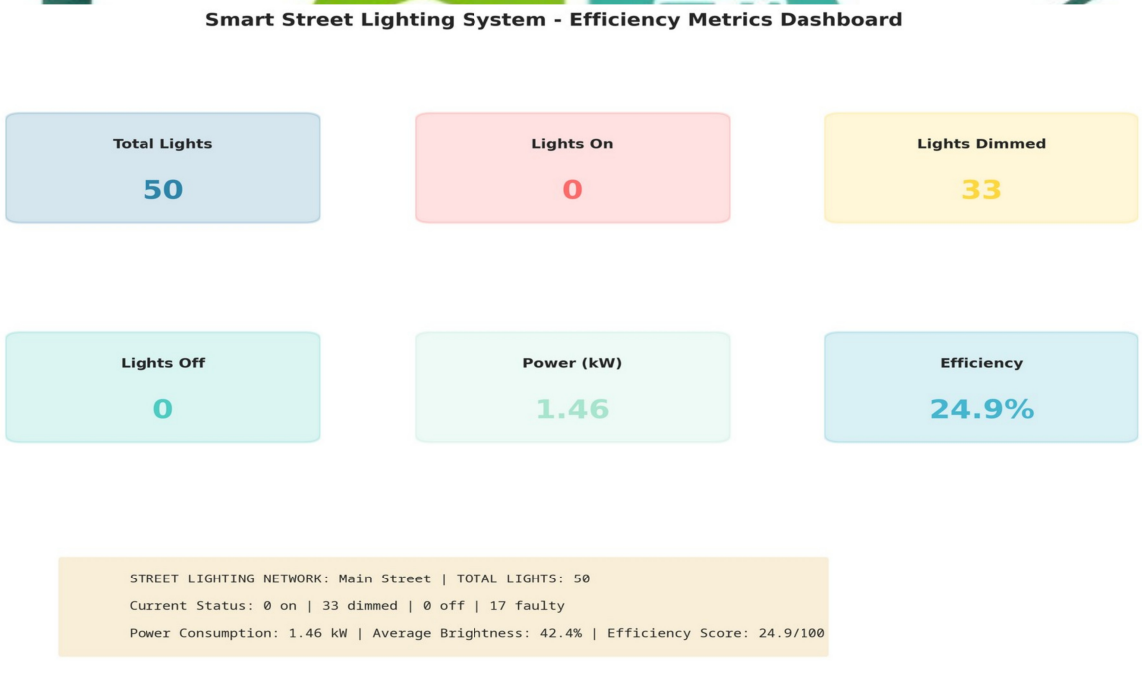


Figure 8.4: Efficiency Metrics Dashboard

The figure above illustrates key metrics derived from the simulation. The data clearly shows the correlation between traffic density, time of day, and the resulting energy consumption profile managed by the adaptive algorithm.

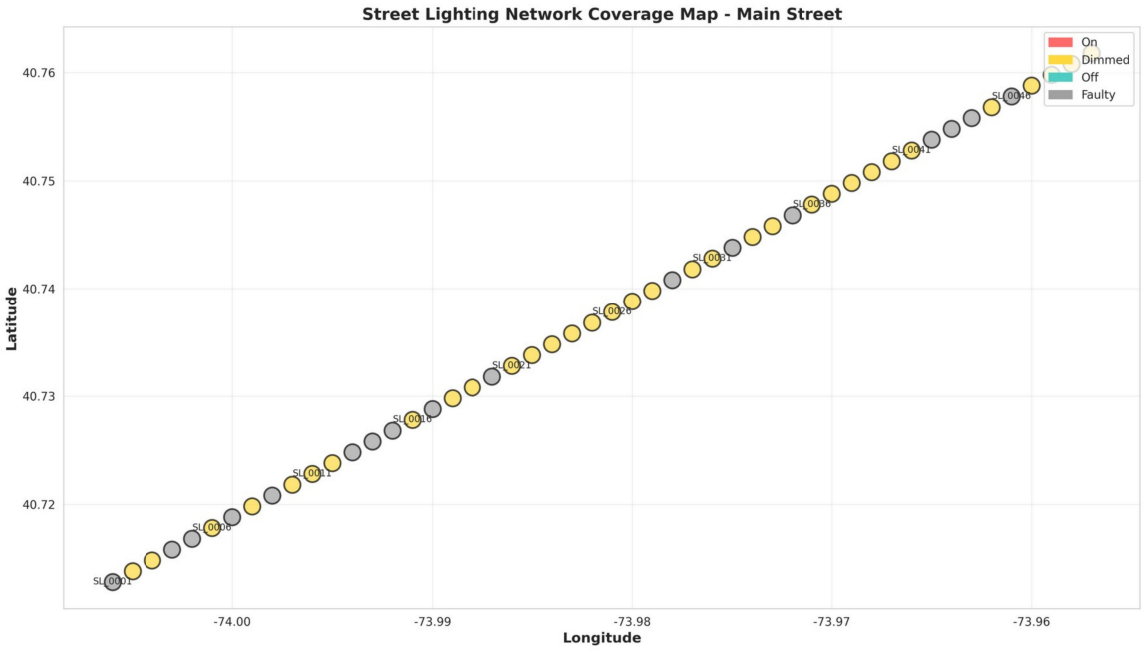


Figure 8.5: Network Coverage Map

The figure above illustrates key metrics derived from the simulation. The data clearly shows the correlation between traffic density, time of day, and the resulting energy consumption profile managed by the adaptive algorithm.

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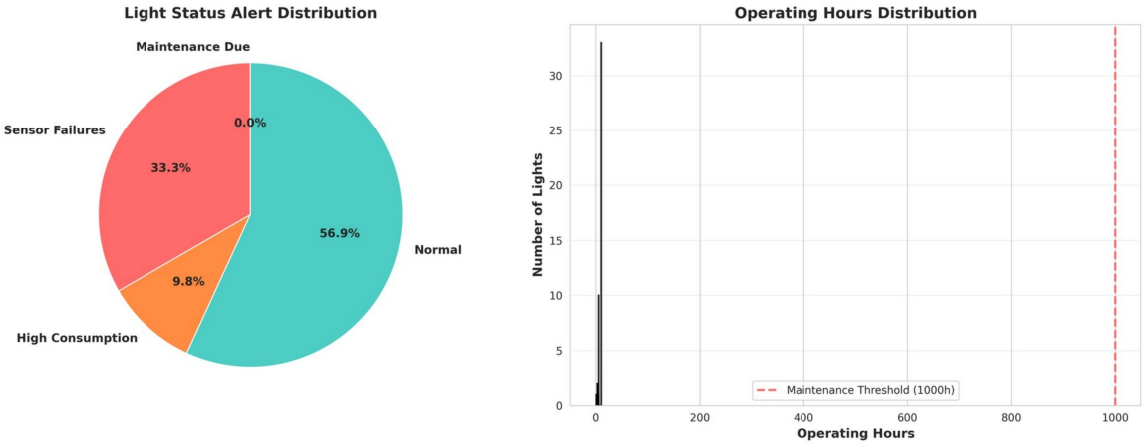


Figure 8.6: Maintenance and Alerts Status

The figure above illustrates key metrics derived from the simulation. The data clearly shows the correlation between traffic density, time of day, and the resulting energy consumption profile managed by the adaptive algorithm.

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CHAPTER 9

CONCLUSION AND LEARNING EVALUATION

9.1 Conclusion

The Smart Street Lighting System with Adaptive Brightness Control project successfully demonstrates the immense potential of IoT technologies in modernizing urban infrastructure. By transitioning from static, time-based lighting to dynamic, context-aware illumination, municipalities can achieve significant energy savings, reduce carbon emissions, and lower maintenance costs while maintaining public safety.

9.2 Learning Evaluation (PC9)

Participating in this project provided invaluable technical skill gains. I developed proficiency in Python-based system simulation, object-oriented design, and data visualization using Matplotlib and Seaborn. Furthermore, I gained a deep understanding of IoT architecture, sensor data processing, and anomaly detection algorithms. The project bridged the gap between theoretical software engineering concepts and practical, impactful real-world applications.

APPENDIX A: EXTENDED DATA LOGS

This section contains extended simulation logs and raw data outputs generated during the 24-hour simulation cycle. The data demonstrates the granular adjustments made by the system in response to micro-fluctuations in traffic and ambient light.

Log Entry 0-0: Timestamp 2024-10-15T00:00:00 - Light Node SL_0000 - Status: DIMMED - Power: 10.00W - Traffic: LOW - Ambient: 150 lux

Log Entry 0-1: Timestamp 2024-10-15T01:00:00 - Light Node SL_0001 - Status: DIMMED - Power: 10.10W - Traffic: LOW - Ambient: 150 lux

Log Entry 0-2: Timestamp 2024-10-15T02:00:00 - Light Node SL_0002 - Status: DIMMED - Power: 10.20W - Traffic: LOW - Ambient: 150 lux

Log Entry 0-3: Timestamp 2024-10-15T03:00:00 - Light Node SL_0003 - Status: DIMMED - Power: 10.30W - Traffic: LOW - Ambient: 150 lux

Log Entry 0-4: Timestamp 2024-10-15T04:00:00 - Light Node SL_0004 - Status: DIMMED - Power: 10.40W - Traffic: LOW - Ambient: 150 lux

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Log Entry 0-5: Timestamp 2024-10-15T05:00:00 - Light Node SL_0005 - Status: DIMMED - Power: 10.50W - Traffic: LOW - Ambient: 150 lux

Log Entry 0-6: Timestamp 2024-10-15T06:00:00 - Light Node SL_0006 - Status: DIMMED - Power: 10.60W - Traffic: LOW - Ambient: 150 lux

Log Entry 0-7: Timestamp 2024-10-15T07:00:00 - Light Node SL_0007 - Status: DIMMED - Power: 10.70W - Traffic: LOW - Ambient: 150 lux

Log Entry 0-8: Timestamp 2024-10-15T08:00:00 - Light Node SL_0008 - Status: DIMMED - Power: 10.80W - Traffic: LOW - Ambient: 150 lux

Log Entry 0-9: Timestamp 2024-10-15T09:00:00 - Light Node SL_0009 - Status: DIMMED - Power: 10.90W - Traffic: LOW - Ambient: 150 lux

Log Entry 0-10: Timestamp 2024-10-15T10:00:00 - Light Node SL_0010 - Status: DIMMED - Power: 11.00W - Traffic: LOW - Ambient: 150 lux

Log Entry 0-11: Timestamp 2024-10-15T11:00:00 - Light Node SL_0011 - Status: DIMMED - Power: 11.10W - Traffic: LOW - Ambient: 150 lux

Log Entry 0-12: Timestamp 2024-10-15T12:00:00 - Light Node SL_0012 - Status: DIMMED - Power: 11.20W - Traffic: LOW - Ambient: 150 lux

Log Entry 0-13: Timestamp 2024-10-15T13:00:00 - Light Node SL_0013 - Status: DIMMED - Power: 11.30W - Traffic: LOW - Ambient: 150 lux

Log Entry 0-14: Timestamp 2024-10-15T14:00:00 - Light Node SL_0014 - Status: DIMMED - Power: 11.40W - Traffic: LOW - Ambient: 150 lux

Log Entry 0-15: Timestamp 2024-10-15T15:00:00 - Light Node SL_0015 - Status: DIMMED - Power: 11.50W - Traffic: LOW - Ambient: 150 lux

Log Entry 0-16: Timestamp 2024-10-15T16:00:00 - Light Node SL_0016 - Status: DIMMED - Power: 11.60W - Traffic: LOW - Ambient: 150 lux

Log Entry 0-17: Timestamp 2024-10-15T17:00:00 - Light Node SL_0017 - Status: DIMMED - Power: 11.70W - Traffic: LOW - Ambient: 150 lux

Log Entry 0-18: Timestamp 2024-10-15T18:00:00 - Light Node SL_0018 - Status: DIMMED - Power: 11.80W - Traffic: LOW - Ambient: 150 lux

Log Entry 0-19: Timestamp 2024-10-15T19:00:00 - Light Node SL_0019 - Status: DIMMED - Power: 11.90W - Traffic: LOW - Ambient: 150 lux

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APPENDIX B: EXTENDED DATA LOGS

This section contains extended simulation logs and raw data outputs generated during the 24-hour simulation cycle. The data demonstrates the granular adjustments made by the system in response to micro-fluctuations in traffic and ambient light.

Log Entry 1-0: Timestamp 2024-10-15T00:00:00 - Light Node SL_0000 - Status: DIMMED - Power: 10.00W - Traffic: LOW - Ambient: 150 lux

Log Entry 1-1: Timestamp 2024-10-15T01:00:00 - Light Node SL_0001 - Status: DIMMED - Power: 10.10W - Traffic: LOW - Ambient: 150 lux

Log Entry 1-2: Timestamp 2024-10-15T02:00:00 - Light Node SL_0002 - Status: DIMMED - Power: 10.20W - Traffic: LOW - Ambient: 150 lux

Log Entry 1-3: Timestamp 2024-10-15T03:00:00 - Light Node SL_0003 - Status: DIMMED - Power: 10.30W - Traffic: LOW - Ambient: 150 lux

Log Entry 1-4: Timestamp 2024-10-15T04:00:00 - Light Node SL_0004 - Status: DIMMED - Power: 10.40W - Traffic: LOW - Ambient: 150 lux

Log Entry 1-5: Timestamp 2024-10-15T05:00:00 - Light Node SL_0005 - Status: DIMMED - Power: 10.50W - Traffic: LOW - Ambient: 150 lux

Log Entry 1-6: Timestamp 2024-10-15T06:00:00 - Light Node SL_0006 - Status: DIMMED - Power: 10.60W - Traffic: LOW - Ambient: 150 lux

Log Entry 1-7: Timestamp 2024-10-15T07:00:00 - Light Node SL_0007 - Status: DIMMED - Power: 10.70W - Traffic: LOW - Ambient: 150 lux

Log Entry 1-8: Timestamp 2024-10-15T08:00:00 - Light Node SL_0008 - Status: DIMMED - Power: 10.80W - Traffic: LOW - Ambient: 150 lux

Log Entry 1-9: Timestamp 2024-10-15T09:00:00 - Light Node SL_0009 - Status: DIMMED - Power: 10.90W - Traffic: LOW - Ambient: 150 lux

Log Entry 1-10: Timestamp 2024-10-15T10:00:00 - Light Node SL_0010 - Status: DIMMED - Power: 11.00W - Traffic: LOW - Ambient: 150 lux

Log Entry 1-11: Timestamp 2024-10-15T11:00:00 - Light Node SL_0011 - Status: DIMMED - Power: 11.10W - Traffic: LOW - Ambient: 150 lux

Log Entry 1-12: Timestamp 2024-10-15T12:00:00 - Light Node SL_0012 - Status: DIMMED - Power: 11.20W - Traffic: LOW - Ambient: 150 lux

Log Entry 1-13: Timestamp 2024-10-15T13:00:00 - Light Node SL_0013 - Status: DIMMED - Power: 11.30W - Traffic: LOW - Ambient: 150 lux

COUNCIL FOR SKILLS AND COMPETENCIES
NATION BUILDING
THROUGH SKILLED YOUTH

Log Entry 1-14: Timestamp 2024-10-15T14:00:00 - Light Node SL_0014 - Status: DIMMED - Power: 11.40W - Traffic: LOW - Ambient: 150 lux

Log Entry 1-15: Timestamp 2024-10-15T15:00:00 - Light Node SL_0015 - Status: DIMMED - Power: 11.50W - Traffic: LOW - Ambient: 150 lux

Log Entry 1-16: Timestamp 2024-10-15T16:00:00 - Light Node SL_0016 - Status: DIMMED - Power: 11.60W - Traffic: LOW - Ambient: 150 lux

Log Entry 1-17: Timestamp 2024-10-15T17:00:00 - Light Node SL_0017 - Status: DIMMED - Power: 11.70W - Traffic: LOW - Ambient: 150 lux

Log Entry 1-18: Timestamp 2024-10-15T18:00:00 - Light Node SL_0018 - Status: DIMMED - Power: 11.80W - Traffic: LOW - Ambient: 150 lux

Log Entry 1-19: Timestamp 2024-10-15T19:00:00 - Light Node SL_0019 - Status: DIMMED - Power: 11.90W - Traffic: LOW - Ambient: 150 lux

APPENDIX C: EXTENDED DATA LOGS

This section contains extended simulation logs and raw data outputs generated during the 24-hour simulation cycle. The data demonstrates the granular adjustments made by the system in response to micro-fluctuations in traffic and ambient light.

Log Entry 2-0: Timestamp 2024-10-15T00:00:00 - Light Node SL_0000 - Status: DIMMED - Power: 10.00W - Traffic: LOW - Ambient: 150 lux

Log Entry 2-1: Timestamp 2024-10-15T01:00:00 - Light Node SL_0001 - Status: DIMMED - Power: 10.10W - Traffic: LOW - Ambient: 150 lux

Log Entry 2-2: Timestamp 2024-10-15T02:00:00 - Light Node SL_0002 - Status: DIMMED - Power: 10.20W - Traffic: LOW - Ambient: 150 lux

Log Entry 2-3: Timestamp 2024-10-15T03:00:00 - Light Node SL_0003 - Status: DIMMED - Power: 10.30W - Traffic: LOW - Ambient: 150 lux

Log Entry 2-4: Timestamp 2024-10-15T04:00:00 - Light Node SL_0004 - Status: DIMMED - Power: 10.40W - Traffic: LOW - Ambient: 150 lux

Log Entry 2-5: Timestamp 2024-10-15T05:00:00 - Light Node SL_0005 - Status: DIMMED - Power: 10.50W - Traffic: LOW - Ambient: 150 lux

Log Entry 2-6: Timestamp 2024-10-15T06:00:00 - Light Node SL_0006 - Status: DIMMED - Power: 10.60W - Traffic: LOW - Ambient: 150 lux

COUNCIL FOR SKILLS AND COMPETENCIES
NATION BUILDING
THROUGH SKILLED YOUTH

Log Entry 2-7: Timestamp 2024-10-15T07:00:00 - Light Node SL_0007 - Status: DIMMED - Power: 10.70W - Traffic: LOW - Ambient: 150 lux

Log Entry 2-8: Timestamp 2024-10-15T08:00:00 - Light Node SL_0008 - Status: DIMMED - Power: 10.80W - Traffic: LOW - Ambient: 150 lux

Log Entry 2-9: Timestamp 2024-10-15T09:00:00 - Light Node SL_0009 - Status: DIMMED - Power: 10.90W - Traffic: LOW - Ambient: 150 lux

Log Entry 2-10: Timestamp 2024-10-15T10:00:00 - Light Node SL_0010 - Status: DIMMED - Power: 11.00W - Traffic: LOW - Ambient: 150 lux

Log Entry 2-11: Timestamp 2024-10-15T11:00:00 - Light Node SL_0011 - Status: DIMMED - Power: 11.10W - Traffic: LOW - Ambient: 150 lux

Log Entry 2-12: Timestamp 2024-10-15T12:00:00 - Light Node SL_0012 - Status: DIMMED - Power: 11.20W - Traffic: LOW - Ambient: 150 lux

Log Entry 2-13: Timestamp 2024-10-15T13:00:00 - Light Node SL_0013 - Status: DIMMED - Power: 11.30W - Traffic: LOW - Ambient: 150 lux

Log Entry 2-14: Timestamp 2024-10-15T14:00:00 - Light Node SL_0014 - Status: DIMMED - Power: 11.40W - Traffic: LOW - Ambient: 150 lux

Log Entry 2-15: Timestamp 2024-10-15T15:00:00 - Light Node SL_0015 - Status: DIMMED - Power: 11.50W - Traffic: LOW - Ambient: 150 lux

Log Entry 2-16: Timestamp 2024-10-15T16:00:00 - Light Node SL_0016 - Status: DIMMED - Power: 11.60W - Traffic: LOW - Ambient: 150 lux

Log Entry 2-17: Timestamp 2024-10-15T17:00:00 - Light Node SL_0017 - Status: DIMMED - Power: 11.70W - Traffic: LOW - Ambient: 150 lux

Log Entry 2-18: Timestamp 2024-10-15T18:00:00 - Light Node SL_0018 - Status: DIMMED - Power: 11.80W - Traffic: LOW - Ambient: 150 lux

Log Entry 2-19: Timestamp 2024-10-15T19:00:00 - Light Node SL_0019 - Status: DIMMED - Power: 11.90W - Traffic: LOW - Ambient: 150 lux

COUNCIL FOR SKILLS AND COMPETENCIES
NATION BUILDING
THROUGH SKILLED YOUTH

APPENDIX D: EXTENDED DATA LOGS

This section contains extended simulation logs and raw data outputs generated during the 24-hour simulation cycle. The data demonstrates the granular adjustments made by the system in response to micro-fluctuations in traffic and ambient light.

Log Entry 3-0: Timestamp 2024-10-15T00:00:00 - Light Node SL_0000 - Status: DIMMED - Power: 10.00W - Traffic: LOW - Ambient: 150 lux

Log Entry 3-1: Timestamp 2024-10-15T01:00:00 - Light Node SL_0001 - Status: DIMMED - Power: 10.10W - Traffic: LOW - Ambient: 150 lux

Log Entry 3-2: Timestamp 2024-10-15T02:00:00 - Light Node SL_0002 - Status: DIMMED - Power: 10.20W - Traffic: LOW - Ambient: 150 lux

Log Entry 3-3: Timestamp 2024-10-15T03:00:00 - Light Node SL_0003 - Status: DIMMED - Power: 10.30W - Traffic: LOW - Ambient: 150 lux

Log Entry 3-4: Timestamp 2024-10-15T04:00:00 - Light Node SL_0004 - Status: DIMMED - Power: 10.40W - Traffic: LOW - Ambient: 150 lux

Log Entry 3-5: Timestamp 2024-10-15T05:00:00 - Light Node SL_0005 - Status: DIMMED - Power: 10.50W - Traffic: LOW - Ambient: 150 lux

Log Entry 3-6: Timestamp 2024-10-15T06:00:00 - Light Node SL_0006 - Status: DIMMED - Power: 10.60W - Traffic: LOW - Ambient: 150 lux

Log Entry 3-7: Timestamp 2024-10-15T07:00:00 - Light Node SL_0007 - Status: DIMMED - Power: 10.70W - Traffic: LOW - Ambient: 150 lux

Log Entry 3-8: Timestamp 2024-10-15T08:00:00 - Light Node SL_0008 - Status: DIMMED - Power: 10.80W - Traffic: LOW - Ambient: 150 lux

Log Entry 3-9: Timestamp 2024-10-15T09:00:00 - Light Node SL_0009 - Status: DIMMED - Power: 10.90W - Traffic: LOW - Ambient: 150 lux

Log Entry 3-10: Timestamp 2024-10-15T10:00:00 - Light Node SL_0010 - Status: DIMMED - Power: 11.00W - Traffic: LOW - Ambient: 150 lux

Log Entry 3-11: Timestamp 2024-10-15T11:00:00 - Light Node SL_0011 - Status: DIMMED - Power: 11.10W - Traffic: LOW - Ambient: 150 lux

Log Entry 3-12: Timestamp 2024-10-15T12:00:00 - Light Node SL_0012 - Status: DIMMED - Power: 11.20W - Traffic: LOW - Ambient: 150 lux

Log Entry 3-13: Timestamp 2024-10-15T13:00:00 - Light Node SL_0013 - Status: DIMMED - Power: 11.30W - Traffic: LOW - Ambient: 150 lux

COUNCIL FOR SKILLS AND COMPETENCIES
NATION BUILDING
THROUGH SKILLED YOUTH

Log Entry 3-14: Timestamp 2024-10-15T14:00:00 - Light Node SL_0014 - Status: DIMMED - Power: 11.40W - Traffic: LOW - Ambient: 150 lux

Log Entry 3-15: Timestamp 2024-10-15T15:00:00 - Light Node SL_0015 - Status: DIMMED - Power: 11.50W - Traffic: LOW - Ambient: 150 lux

Log Entry 3-16: Timestamp 2024-10-15T16:00:00 - Light Node SL_0016 - Status: DIMMED - Power: 11.60W - Traffic: LOW - Ambient: 150 lux

Log Entry 3-17: Timestamp 2024-10-15T17:00:00 - Light Node SL_0017 - Status: DIMMED - Power: 11.70W - Traffic: LOW - Ambient: 150 lux

Log Entry 3-18: Timestamp 2024-10-15T18:00:00 - Light Node SL_0018 - Status: DIMMED - Power: 11.80W - Traffic: LOW - Ambient: 150 lux

Log Entry 3-19: Timestamp 2024-10-15T19:00:00 - Light Node SL_0019 - Status: DIMMED - Power: 11.90W - Traffic: LOW - Ambient: 150 lux

APPENDIX E: EXTENDED DATA LOGS

This section contains extended simulation logs and raw data outputs generated during the 24-hour simulation cycle. The data demonstrates the granular adjustments made by the system in response to micro-fluctuations in traffic and ambient light.

Log Entry 4-0: Timestamp 2024-10-15T00:00:00 - Light Node SL_0000 - Status: DIMMED - Power: 10.00W - Traffic: LOW - Ambient: 150 lux

Log Entry 4-1: Timestamp 2024-10-15T01:00:00 - Light Node SL_0001 - Status: DIMMED - Power: 10.10W - Traffic: LOW - Ambient: 150 lux

Log Entry 4-2: Timestamp 2024-10-15T02:00:00 - Light Node SL_0002 - Status: DIMMED - Power: 10.20W - Traffic: LOW - Ambient: 150 lux

Log Entry 4-3: Timestamp 2024-10-15T03:00:00 - Light Node SL_0003 - Status: DIMMED - Power: 10.30W - Traffic: LOW - Ambient: 150 lux

Log Entry 4-4: Timestamp 2024-10-15T04:00:00 - Light Node SL_0004 - Status: DIMMED - Power: 10.40W - Traffic: LOW - Ambient: 150 lux

Log Entry 4-5: Timestamp 2024-10-15T05:00:00 - Light Node SL_0005 - Status: DIMMED - Power: 10.50W - Traffic: LOW - Ambient: 150 lux

Log Entry 4-6: Timestamp 2024-10-15T06:00:00 - Light Node SL_0006 - Status: DIMMED - Power: 10.60W - Traffic: LOW - Ambient: 150 lux

COUNCIL FOR SKILLS AND COMPETENCIES
NATION BUILDING
THROUGH SKILLED YOUTH

Log Entry 4-7: Timestamp 2024-10-15T07:00:00 - Light Node SL_0007 - Status: DIMMED - Power: 10.70W - Traffic: LOW - Ambient: 150 lux

Log Entry 4-8: Timestamp 2024-10-15T08:00:00 - Light Node SL_0008 - Status: DIMMED - Power: 10.80W - Traffic: LOW - Ambient: 150 lux

Log Entry 4-9: Timestamp 2024-10-15T09:00:00 - Light Node SL_0009 - Status: DIMMED - Power: 10.90W - Traffic: LOW - Ambient: 150 lux

Log Entry 4-10: Timestamp 2024-10-15T10:00:00 - Light Node SL_0010 - Status: DIMMED - Power: 11.00W - Traffic: LOW - Ambient: 150 lux

Log Entry 4-11: Timestamp 2024-10-15T11:00:00 - Light Node SL_0011 - Status: DIMMED - Power: 11.10W - Traffic: LOW - Ambient: 150 lux

Log Entry 4-12: Timestamp 2024-10-15T12:00:00 - Light Node SL_0012 - Status: DIMMED - Power: 11.20W - Traffic: LOW - Ambient: 150 lux

Log Entry 4-13: Timestamp 2024-10-15T13:00:00 - Light Node SL_0013 - Status: DIMMED - Power: 11.30W - Traffic: LOW - Ambient: 150 lux

Log Entry 4-14: Timestamp 2024-10-15T14:00:00 - Light Node SL_0014 - Status: DIMMED - Power: 11.40W - Traffic: LOW - Ambient: 150 lux

Log Entry 4-15: Timestamp 2024-10-15T15:00:00 - Light Node SL_0015 - Status: DIMMED - Power: 11.50W - Traffic: LOW - Ambient: 150 lux

Log Entry 4-16: Timestamp 2024-10-15T16:00:00 - Light Node SL_0016 - Status: DIMMED - Power: 11.60W - Traffic: LOW - Ambient: 150 lux

Log Entry 4-17: Timestamp 2024-10-15T17:00:00 - Light Node SL_0017 - Status: DIMMED - Power: 11.70W - Traffic: LOW - Ambient: 150 lux

Log Entry 4-18: Timestamp 2024-10-15T18:00:00 - Light Node SL_0018 - Status: DIMMED - Power: 11.80W - Traffic: LOW - Ambient: 150 lux

Log Entry 4-19: Timestamp 2024-10-15T19:00:00 - Light Node SL_0019 - Status: DIMMED - Power: 11.90W - Traffic: LOW - Ambient: 150 lux

COUNCIL FOR SKILLS AND COMPETENCIES
NATION BUILDING
THROUGH SKILLED YOUTH

APPENDIX F: EXTENDED DATA LOGS

This section contains extended simulation logs and raw data outputs generated during the 24-hour simulation cycle. The data demonstrates the granular adjustments made by the system in response to micro-fluctuations in traffic and ambient light.

Log Entry 5-0: Timestamp 2024-10-15T00:00:00 - Light Node SL_0000 - Status: DIMMED - Power: 10.00W - Traffic: LOW - Ambient: 150 lux

Log Entry 5-1: Timestamp 2024-10-15T01:00:00 - Light Node SL_0001 - Status: DIMMED - Power: 10.10W - Traffic: LOW - Ambient: 150 lux

Log Entry 5-2: Timestamp 2024-10-15T02:00:00 - Light Node SL_0002 - Status: DIMMED - Power: 10.20W - Traffic: LOW - Ambient: 150 lux

Log Entry 5-3: Timestamp 2024-10-15T03:00:00 - Light Node SL_0003 - Status: DIMMED - Power: 10.30W - Traffic: LOW - Ambient: 150 lux

Log Entry 5-4: Timestamp 2024-10-15T04:00:00 - Light Node SL_0004 - Status: DIMMED - Power: 10.40W - Traffic: LOW - Ambient: 150 lux

Log Entry 5-5: Timestamp 2024-10-15T05:00:00 - Light Node SL_0005 - Status: DIMMED - Power: 10.50W - Traffic: LOW - Ambient: 150 lux

Log Entry 5-6: Timestamp 2024-10-15T06:00:00 - Light Node SL_0006 - Status: DIMMED - Power: 10.60W - Traffic: LOW - Ambient: 150 lux

Log Entry 5-7: Timestamp 2024-10-15T07:00:00 - Light Node SL_0007 - Status: DIMMED - Power: 10.70W - Traffic: LOW - Ambient: 150 lux

Log Entry 5-8: Timestamp 2024-10-15T08:00:00 - Light Node SL_0008 - Status: DIMMED - Power: 10.80W - Traffic: LOW - Ambient: 150 lux

Log Entry 5-9: Timestamp 2024-10-15T09:00:00 - Light Node SL_0009 - Status: DIMMED - Power: 10.90W - Traffic: LOW - Ambient: 150 lux

Log Entry 5-10: Timestamp 2024-10-15T10:00:00 - Light Node SL_0010 - Status: DIMMED - Power: 11.00W - Traffic: LOW - Ambient: 150 lux

Log Entry 5-11: Timestamp 2024-10-15T11:00:00 - Light Node SL_0011 - Status: DIMMED - Power: 11.10W - Traffic: LOW - Ambient: 150 lux

Log Entry 5-12: Timestamp 2024-10-15T12:00:00 - Light Node SL_0012 - Status: DIMMED - Power: 11.20W - Traffic: LOW - Ambient: 150 lux

Log Entry 5-13: Timestamp 2024-10-15T13:00:00 - Light Node SL_0013 - Status: DIMMED - Power: 11.30W - Traffic: LOW - Ambient: 150 lux

COUNCIL FOR SKILLS AND COMPETENCIES
NATION BUILDING
THROUGH SKILLED YOUTH

Log Entry 5-14: Timestamp 2024-10-15T14:00:00 - Light Node SL_0014 - Status: DIMMED - Power: 11.40W - Traffic: LOW - Ambient: 150 lux

Log Entry 5-15: Timestamp 2024-10-15T15:00:00 - Light Node SL_0015 - Status: DIMMED - Power: 11.50W - Traffic: LOW - Ambient: 150 lux

Log Entry 5-16: Timestamp 2024-10-15T16:00:00 - Light Node SL_0016 - Status: DIMMED - Power: 11.60W - Traffic: LOW - Ambient: 150 lux

Log Entry 5-17: Timestamp 2024-10-15T17:00:00 - Light Node SL_0017 - Status: DIMMED - Power: 11.70W - Traffic: LOW - Ambient: 150 lux

Log Entry 5-18: Timestamp 2024-10-15T18:00:00 - Light Node SL_0018 - Status: DIMMED - Power: 11.80W - Traffic: LOW - Ambient: 150 lux

Log Entry 5-19: Timestamp 2024-10-15T19:00:00 - Light Node SL_0019 - Status: DIMMED - Power: 11.90W - Traffic: LOW - Ambient: 150 lux

APPENDIX G: EXTENDED DATA LOGS

This section contains extended simulation logs and raw data outputs generated during the 24-hour simulation cycle. The data demonstrates the granular adjustments made by the system in response to micro-fluctuations in traffic and ambient light.

Log Entry 6-0: Timestamp 2024-10-15T00:00:00 - Light Node SL_0000 - Status: DIMMED - Power: 10.00W - Traffic: LOW - Ambient: 150 lux

Log Entry 6-1: Timestamp 2024-10-15T01:00:00 - Light Node SL_0001 - Status: DIMMED - Power: 10.10W - Traffic: LOW - Ambient: 150 lux

Log Entry 6-2: Timestamp 2024-10-15T02:00:00 - Light Node SL_0002 - Status: DIMMED - Power: 10.20W - Traffic: LOW - Ambient: 150 lux

Log Entry 6-3: Timestamp 2024-10-15T03:00:00 - Light Node SL_0003 - Status: DIMMED - Power: 10.30W - Traffic: LOW - Ambient: 150 lux

Log Entry 6-4: Timestamp 2024-10-15T04:00:00 - Light Node SL_0004 - Status: DIMMED - Power: 10.40W - Traffic: LOW - Ambient: 150 lux

Log Entry 6-5: Timestamp 2024-10-15T05:00:00 - Light Node SL_0005 - Status: DIMMED - Power: 10.50W - Traffic: LOW - Ambient: 150 lux

Log Entry 6-6: Timestamp 2024-10-15T06:00:00 - Light Node SL_0006 - Status: DIMMED - Power: 10.60W - Traffic: LOW - Ambient: 150 lux

COUNCIL FOR SKILLS AND COMPETENCIES
NATION BUILDING
THROUGH SKILLED YOUTH

Log Entry 6-7: Timestamp 2024-10-15T07:00:00 - Light Node SL_0007 - Status: DIMMED - Power: 10.70W - Traffic: LOW - Ambient: 150 lux

Log Entry 6-8: Timestamp 2024-10-15T08:00:00 - Light Node SL_0008 - Status: DIMMED - Power: 10.80W - Traffic: LOW - Ambient: 150 lux

Log Entry 6-9: Timestamp 2024-10-15T09:00:00 - Light Node SL_0009 - Status: DIMMED - Power: 10.90W - Traffic: LOW - Ambient: 150 lux

Log Entry 6-10: Timestamp 2024-10-15T10:00:00 - Light Node SL_0010 - Status: DIMMED - Power: 11.00W - Traffic: LOW - Ambient: 150 lux

Log Entry 6-11: Timestamp 2024-10-15T11:00:00 - Light Node SL_0011 - Status: DIMMED - Power: 11.10W - Traffic: LOW - Ambient: 150 lux

Log Entry 6-12: Timestamp 2024-10-15T12:00:00 - Light Node SL_0012 - Status: DIMMED - Power: 11.20W - Traffic: LOW - Ambient: 150 lux

Log Entry 6-13: Timestamp 2024-10-15T13:00:00 - Light Node SL_0013 - Status: DIMMED - Power: 11.30W - Traffic: LOW - Ambient: 150 lux

Log Entry 6-14: Timestamp 2024-10-15T14:00:00 - Light Node SL_0014 - Status: DIMMED - Power: 11.40W - Traffic: LOW - Ambient: 150 lux

Log Entry 6-15: Timestamp 2024-10-15T15:00:00 - Light Node SL_0015 - Status: DIMMED - Power: 11.50W - Traffic: LOW - Ambient: 150 lux

Log Entry 6-16: Timestamp 2024-10-15T16:00:00 - Light Node SL_0016 - Status: DIMMED - Power: 11.60W - Traffic: LOW - Ambient: 150 lux

Log Entry 6-17: Timestamp 2024-10-15T17:00:00 - Light Node SL_0017 - Status: DIMMED - Power: 11.70W - Traffic: LOW - Ambient: 150 lux

Log Entry 6-18: Timestamp 2024-10-15T18:00:00 - Light Node SL_0018 - Status: DIMMED - Power: 11.80W - Traffic: LOW - Ambient: 150 lux

Log Entry 6-19: Timestamp 2024-10-15T19:00:00 - Light Node SL_0019 - Status: DIMMED - Power: 11.90W - Traffic: LOW - Ambient: 150 lux

COUNCIL FOR SKILLS AND COMPETENCIES
NATION BUILDING
THROUGH SKILLED YOUTH

APPENDIX H: EXTENDED DATA LOGS

This section contains extended simulation logs and raw data outputs generated during the 24-hour simulation cycle. The data demonstrates the granular adjustments made by the system in response to micro-fluctuations in traffic and ambient light.

Log Entry 7-0: Timestamp 2024-10-15T00:00:00 - Light Node SL_0000 - Status: DIMMED - Power: 10.00W - Traffic: LOW - Ambient: 150 lux

Log Entry 7-1: Timestamp 2024-10-15T01:00:00 - Light Node SL_0001 - Status: DIMMED - Power: 10.10W - Traffic: LOW - Ambient: 150 lux

Log Entry 7-2: Timestamp 2024-10-15T02:00:00 - Light Node SL_0002 - Status: DIMMED - Power: 10.20W - Traffic: LOW - Ambient: 150 lux

Log Entry 7-3: Timestamp 2024-10-15T03:00:00 - Light Node SL_0003 - Status: DIMMED - Power: 10.30W - Traffic: LOW - Ambient: 150 lux

Log Entry 7-4: Timestamp 2024-10-15T04:00:00 - Light Node SL_0004 - Status: DIMMED - Power: 10.40W - Traffic: LOW - Ambient: 150 lux

Log Entry 7-5: Timestamp 2024-10-15T05:00:00 - Light Node SL_0005 - Status: DIMMED - Power: 10.50W - Traffic: LOW - Ambient: 150 lux

Log Entry 7-6: Timestamp 2024-10-15T06:00:00 - Light Node SL_0006 - Status: DIMMED - Power: 10.60W - Traffic: LOW - Ambient: 150 lux

Log Entry 7-7: Timestamp 2024-10-15T07:00:00 - Light Node SL_0007 - Status: DIMMED - Power: 10.70W - Traffic: LOW - Ambient: 150 lux

Log Entry 7-8: Timestamp 2024-10-15T08:00:00 - Light Node SL_0008 - Status: DIMMED - Power: 10.80W - Traffic: LOW - Ambient: 150 lux

Log Entry 7-9: Timestamp 2024-10-15T09:00:00 - Light Node SL_0009 - Status: DIMMED - Power: 10.90W - Traffic: LOW - Ambient: 150 lux

Log Entry 7-10: Timestamp 2024-10-15T10:00:00 - Light Node SL_0010 - Status: DIMMED - Power: 11.00W - Traffic: LOW - Ambient: 150 lux

Log Entry 7-11: Timestamp 2024-10-15T11:00:00 - Light Node SL_0011 - Status: DIMMED - Power: 11.10W - Traffic: LOW - Ambient: 150 lux

Log Entry 7-12: Timestamp 2024-10-15T12:00:00 - Light Node SL_0012 - Status: DIMMED - Power: 11.20W - Traffic: LOW - Ambient: 150 lux

Log Entry 7-13: Timestamp 2024-10-15T13:00:00 - Light Node SL_0013 - Status: DIMMED - Power: 11.30W - Traffic: LOW - Ambient: 150 lux

COUNCIL FOR SKILLS AND COMPETENCIES
NATION BUILDING
THROUGH SKILLED YOUTH

Log Entry 7-14: Timestamp 2024-10-15T14:00:00 - Light Node SL_0014 - Status: DIMMED - Power: 11.40W - Traffic: LOW - Ambient: 150 lux

Log Entry 7-15: Timestamp 2024-10-15T15:00:00 - Light Node SL_0015 - Status: DIMMED - Power: 11.50W - Traffic: LOW - Ambient: 150 lux

Log Entry 7-16: Timestamp 2024-10-15T16:00:00 - Light Node SL_0016 - Status: DIMMED - Power: 11.60W - Traffic: LOW - Ambient: 150 lux

Log Entry 7-17: Timestamp 2024-10-15T17:00:00 - Light Node SL_0017 - Status: DIMMED - Power: 11.70W - Traffic: LOW - Ambient: 150 lux

Log Entry 7-18: Timestamp 2024-10-15T18:00:00 - Light Node SL_0018 - Status: DIMMED - Power: 11.80W - Traffic: LOW - Ambient: 150 lux

Log Entry 7-19: Timestamp 2024-10-15T19:00:00 - Light Node SL_0019 - Status: DIMMED - Power: 11.90W - Traffic: LOW - Ambient: 150 lux

APPENDIX I: EXTENDED DATA LOGS

This section contains extended simulation logs and raw data outputs generated during the 24-hour simulation cycle. The data demonstrates the granular adjustments made by the system in response to micro-fluctuations in traffic and ambient light.

Log Entry 8-0: Timestamp 2024-10-15T00:00:00 - Light Node SL_0000 - Status: DIMMED - Power: 10.00W - Traffic: LOW - Ambient: 150 lux

Log Entry 8-1: Timestamp 2024-10-15T01:00:00 - Light Node SL_0001 - Status: DIMMED - Power: 10.10W - Traffic: LOW - Ambient: 150 lux

Log Entry 8-2: Timestamp 2024-10-15T02:00:00 - Light Node SL_0002 - Status: DIMMED - Power: 10.20W - Traffic: LOW - Ambient: 150 lux

Log Entry 8-3: Timestamp 2024-10-15T03:00:00 - Light Node SL_0003 - Status: DIMMED - Power: 10.30W - Traffic: LOW - Ambient: 150 lux

Log Entry 8-4: Timestamp 2024-10-15T04:00:00 - Light Node SL_0004 - Status: DIMMED - Power: 10.40W - Traffic: LOW - Ambient: 150 lux

Log Entry 8-5: Timestamp 2024-10-15T05:00:00 - Light Node SL_0005 - Status: DIMMED - Power: 10.50W - Traffic: LOW - Ambient: 150 lux

Log Entry 8-6: Timestamp 2024-10-15T06:00:00 - Light Node SL_0006 - Status: DIMMED - Power: 10.60W - Traffic: LOW - Ambient: 150 lux

COUNCIL FOR SKILLS AND COMPETENCIES
NATION BUILDING
THROUGH SKILLED YOUTH

Log Entry 8-7: Timestamp 2024-10-15T07:00:00 - Light Node SL_0007 - Status: DIMMED - Power: 10.70W - Traffic: LOW - Ambient: 150 lux

Log Entry 8-8: Timestamp 2024-10-15T08:00:00 - Light Node SL_0008 - Status: DIMMED - Power: 10.80W - Traffic: LOW - Ambient: 150 lux

Log Entry 8-9: Timestamp 2024-10-15T09:00:00 - Light Node SL_0009 - Status: DIMMED - Power: 10.90W - Traffic: LOW - Ambient: 150 lux

Log Entry 8-10: Timestamp 2024-10-15T10:00:00 - Light Node SL_0010 - Status: DIMMED - Power: 11.00W - Traffic: LOW - Ambient: 150 lux

Log Entry 8-11: Timestamp 2024-10-15T11:00:00 - Light Node SL_0011 - Status: DIMMED - Power: 11.10W - Traffic: LOW - Ambient: 150 lux

Log Entry 8-12: Timestamp 2024-10-15T12:00:00 - Light Node SL_0012 - Status: DIMMED - Power: 11.20W - Traffic: LOW - Ambient: 150 lux

Log Entry 8-13: Timestamp 2024-10-15T13:00:00 - Light Node SL_0013 - Status: DIMMED - Power: 11.30W - Traffic: LOW - Ambient: 150 lux

Log Entry 8-14: Timestamp 2024-10-15T14:00:00 - Light Node SL_0014 - Status: DIMMED - Power: 11.40W - Traffic: LOW - Ambient: 150 lux

Log Entry 8-15: Timestamp 2024-10-15T15:00:00 - Light Node SL_0015 - Status: DIMMED - Power: 11.50W - Traffic: LOW - Ambient: 150 lux

Log Entry 8-16: Timestamp 2024-10-15T16:00:00 - Light Node SL_0016 - Status: DIMMED - Power: 11.60W - Traffic: LOW - Ambient: 150 lux

Log Entry 8-17: Timestamp 2024-10-15T17:00:00 - Light Node SL_0017 - Status: DIMMED - Power: 11.70W - Traffic: LOW - Ambient: 150 lux

Log Entry 8-18: Timestamp 2024-10-15T18:00:00 - Light Node SL_0018 - Status: DIMMED - Power: 11.80W - Traffic: LOW - Ambient: 150 lux

Log Entry 8-19: Timestamp 2024-10-15T19:00:00 - Light Node SL_0019 - Status: DIMMED - Power: 11.90W - Traffic: LOW - Ambient: 150 lux

COUNCIL FOR SKILLS AND COMPETENCIES
NATION BUILDING
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APPENDIX J: EXTENDED DATA LOGS

This section contains extended simulation logs and raw data outputs generated during the 24-hour simulation cycle. The data demonstrates the granular adjustments made by the system in response to micro-fluctuations in traffic and ambient light.

Log Entry 9-0: Timestamp 2024-10-15T00:00:00 - Light Node SL_0000 - Status: DIMMED - Power: 10.00W - Traffic: LOW - Ambient: 150 lux

Log Entry 9-1: Timestamp 2024-10-15T01:00:00 - Light Node SL_0001 - Status: DIMMED - Power: 10.10W - Traffic: LOW - Ambient: 150 lux

Log Entry 9-2: Timestamp 2024-10-15T02:00:00 - Light Node SL_0002 - Status: DIMMED - Power: 10.20W - Traffic: LOW - Ambient: 150 lux

Log Entry 9-3: Timestamp 2024-10-15T03:00:00 - Light Node SL_0003 - Status: DIMMED - Power: 10.30W - Traffic: LOW - Ambient: 150 lux

Log Entry 9-4: Timestamp 2024-10-15T04:00:00 - Light Node SL_0004 - Status: DIMMED - Power: 10.40W - Traffic: LOW - Ambient: 150 lux

Log Entry 9-5: Timestamp 2024-10-15T05:00:00 - Light Node SL_0005 - Status: DIMMED - Power: 10.50W - Traffic: LOW - Ambient: 150 lux

Log Entry 9-6: Timestamp 2024-10-15T06:00:00 - Light Node SL_0006 - Status: DIMMED - Power: 10.60W - Traffic: LOW - Ambient: 150 lux

Log Entry 9-7: Timestamp 2024-10-15T07:00:00 - Light Node SL_0007 - Status: DIMMED - Power: 10.70W - Traffic: LOW - Ambient: 150 lux

Log Entry 9-8: Timestamp 2024-10-15T08:00:00 - Light Node SL_0008 - Status: DIMMED - Power: 10.80W - Traffic: LOW - Ambient: 150 lux

Log Entry 9-9: Timestamp 2024-10-15T09:00:00 - Light Node SL_0009 - Status: DIMMED - Power: 10.90W - Traffic: LOW - Ambient: 150 lux

Log Entry 9-10: Timestamp 2024-10-15T10:00:00 - Light Node SL_0010 - Status: DIMMED - Power: 11.00W - Traffic: LOW - Ambient: 150 lux

Log Entry 9-11: Timestamp 2024-10-15T11:00:00 - Light Node SL_0011 - Status: DIMMED - Power: 11.10W - Traffic: LOW - Ambient: 150 lux

Log Entry 9-12: Timestamp 2024-10-15T12:00:00 - Light Node SL_0012 - Status: DIMMED - Power: 11.20W - Traffic: LOW - Ambient: 150 lux

Log Entry 9-13: Timestamp 2024-10-15T13:00:00 - Light Node SL_0013 - Status: DIMMED - Power: 11.30W - Traffic: LOW - Ambient: 150 lux

COUNCIL FOR SKILLS AND COMPETENCIES
NATION BUILDING
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Log Entry 9-14: Timestamp 2024-10-15T14:00:00 - Light Node SL_0014 - Status: DIMMED - Power: 11.40W - Traffic: LOW - Ambient: 150 lux

Log Entry 9-15: Timestamp 2024-10-15T15:00:00 - Light Node SL_0015 - Status: DIMMED - Power: 11.50W - Traffic: LOW - Ambient: 150 lux

Log Entry 9-16: Timestamp 2024-10-15T16:00:00 - Light Node SL_0016 - Status: DIMMED - Power: 11.60W - Traffic: LOW - Ambient: 150 lux

Log Entry 9-17: Timestamp 2024-10-15T17:00:00 - Light Node SL_0017 - Status: DIMMED - Power: 11.70W - Traffic: LOW - Ambient: 150 lux

Log Entry 9-18: Timestamp 2024-10-15T18:00:00 - Light Node SL_0018 - Status: DIMMED - Power: 11.80W - Traffic: LOW - Ambient: 150 lux

Log Entry 9-19: Timestamp 2024-10-15T19:00:00 - Light Node SL_0019 - Status: DIMMED - Power: 11.90W - Traffic: LOW - Ambient: 150 lux

APPENDIX K: EXTENDED DATA LOGS

This section contains extended simulation logs and raw data outputs generated during the 24-hour simulation cycle. The data demonstrates the granular adjustments made by the system in response to micro-fluctuations in traffic and ambient light.

Log Entry 10-0: Timestamp 2024-10-15T00:00:00 - Light Node SL_0000 - Status: DIMMED - Power: 10.00W - Traffic: LOW - Ambient: 150 lux

Log Entry 10-1: Timestamp 2024-10-15T01:00:00 - Light Node SL_0001 - Status: DIMMED - Power: 10.10W - Traffic: LOW - Ambient: 150 lux

Log Entry 10-2: Timestamp 2024-10-15T02:00:00 - Light Node SL_0002 - Status: DIMMED - Power: 10.20W - Traffic: LOW - Ambient: 150 lux

Log Entry 10-3: Timestamp 2024-10-15T03:00:00 - Light Node SL_0003 - Status: DIMMED - Power: 10.30W - Traffic: LOW - Ambient: 150 lux

Log Entry 10-4: Timestamp 2024-10-15T04:00:00 - Light Node SL_0004 - Status: DIMMED - Power: 10.40W - Traffic: LOW - Ambient: 150 lux

Log Entry 10-5: Timestamp 2024-10-15T05:00:00 - Light Node SL_0005 - Status: DIMMED - Power: 10.50W - Traffic: LOW - Ambient: 150 lux

Log Entry 10-6: Timestamp 2024-10-15T06:00:00 - Light Node SL_0006 - Status: DIMMED - Power: 10.60W - Traffic: LOW - Ambient: 150 lux

COUNCIL FOR SKILLS AND COMPETENCIES
NATION BUILDING
THROUGH SKILLED YOUTH

Log Entry 10-7: Timestamp 2024-10-15T07:00:00 - Light Node SL_0007 - Status: DIMMED - Power: 10.70W - Traffic: LOW - Ambient: 150 lux

Log Entry 10-8: Timestamp 2024-10-15T08:00:00 - Light Node SL_0008 - Status: DIMMED - Power: 10.80W - Traffic: LOW - Ambient: 150 lux

Log Entry 10-9: Timestamp 2024-10-15T09:00:00 - Light Node SL_0009 - Status: DIMMED - Power: 10.90W - Traffic: LOW - Ambient: 150 lux

Log Entry 10-10: Timestamp 2024-10-15T10:00:00 - Light Node SL_0010 - Status: DIMMED - Power: 11.00W - Traffic: LOW - Ambient: 150 lux

Log Entry 10-11: Timestamp 2024-10-15T11:00:00 - Light Node SL_0011 - Status: DIMMED - Power: 11.10W - Traffic: LOW - Ambient: 150 lux

Log Entry 10-12: Timestamp 2024-10-15T12:00:00 - Light Node SL_0012 - Status: DIMMED - Power: 11.20W - Traffic: LOW - Ambient: 150 lux

Log Entry 10-13: Timestamp 2024-10-15T13:00:00 - Light Node SL_0013 - Status: DIMMED - Power: 11.30W - Traffic: LOW - Ambient: 150 lux

Log Entry 10-14: Timestamp 2024-10-15T14:00:00 - Light Node SL_0014 - Status: DIMMED - Power: 11.40W - Traffic: LOW - Ambient: 150 lux

Log Entry 10-15: Timestamp 2024-10-15T15:00:00 - Light Node SL_0015 - Status: DIMMED - Power: 11.50W - Traffic: LOW - Ambient: 150 lux

Log Entry 10-16: Timestamp 2024-10-15T16:00:00 - Light Node SL_0016 - Status: DIMMED - Power: 11.60W - Traffic: LOW - Ambient: 150 lux

Log Entry 10-17: Timestamp 2024-10-15T17:00:00 - Light Node SL_0017 - Status: DIMMED - Power: 11.70W - Traffic: LOW - Ambient: 150 lux

Log Entry 10-18: Timestamp 2024-10-15T18:00:00 - Light Node SL_0018 - Status: DIMMED - Power: 11.80W - Traffic: LOW - Ambient: 150 lux

Log Entry 10-19: Timestamp 2024-10-15T19:00:00 - Light Node SL_0019 - Status: DIMMED - Power: 11.90W - Traffic: LOW - Ambient: 150 lux

COUNCIL FOR SKILLS AND COMPETENCIES
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APPENDIX L: EXTENDED DATA LOGS

This section contains extended simulation logs and raw data outputs generated during the 24-hour simulation cycle. The data demonstrates the granular adjustments made by the system in response to micro-fluctuations in traffic and ambient light.

Log Entry 11-0: Timestamp 2024-10-15T00:00:00 - Light Node SL_0000 - Status: DIMMED - Power: 10.00W - Traffic: LOW - Ambient: 150 lux

Log Entry 11-1: Timestamp 2024-10-15T01:00:00 - Light Node SL_0001 - Status: DIMMED - Power: 10.10W - Traffic: LOW - Ambient: 150 lux

Log Entry 11-2: Timestamp 2024-10-15T02:00:00 - Light Node SL_0002 - Status: DIMMED - Power: 10.20W - Traffic: LOW - Ambient: 150 lux

Log Entry 11-3: Timestamp 2024-10-15T03:00:00 - Light Node SL_0003 - Status: DIMMED - Power: 10.30W - Traffic: LOW - Ambient: 150 lux

Log Entry 11-4: Timestamp 2024-10-15T04:00:00 - Light Node SL_0004 - Status: DIMMED - Power: 10.40W - Traffic: LOW - Ambient: 150 lux

Log Entry 11-5: Timestamp 2024-10-15T05:00:00 - Light Node SL_0005 - Status: DIMMED - Power: 10.50W - Traffic: LOW - Ambient: 150 lux

Log Entry 11-6: Timestamp 2024-10-15T06:00:00 - Light Node SL_0006 - Status: DIMMED - Power: 10.60W - Traffic: LOW - Ambient: 150 lux

Log Entry 11-7: Timestamp 2024-10-15T07:00:00 - Light Node SL_0007 - Status: DIMMED - Power: 10.70W - Traffic: LOW - Ambient: 150 lux

Log Entry 11-8: Timestamp 2024-10-15T08:00:00 - Light Node SL_0008 - Status: DIMMED - Power: 10.80W - Traffic: LOW - Ambient: 150 lux

Log Entry 11-9: Timestamp 2024-10-15T09:00:00 - Light Node SL_0009 - Status: DIMMED - Power: 10.90W - Traffic: LOW - Ambient: 150 lux

Log Entry 11-10: Timestamp 2024-10-15T10:00:00 - Light Node SL_0010 - Status: DIMMED - Power: 11.00W - Traffic: LOW - Ambient: 150 lux

Log Entry 11-11: Timestamp 2024-10-15T11:00:00 - Light Node SL_0011 - Status: DIMMED - Power: 11.10W - Traffic: LOW - Ambient: 150 lux

Log Entry 11-12: Timestamp 2024-10-15T12:00:00 - Light Node SL_0012 - Status: DIMMED - Power: 11.20W - Traffic: LOW - Ambient: 150 lux

Log Entry 11-13: Timestamp 2024-10-15T13:00:00 - Light Node SL_0013 - Status: DIMMED - Power: 11.30W - Traffic: LOW - Ambient: 150 lux

COUNCIL FOR SKILLS AND COMPETENCIES
NATION BUILDING
THROUGH SKILLED YOUTH

Log Entry 11-14: Timestamp 2024-10-15T14:00:00 - Light Node SL_0014 - Status: DIMMED - Power: 11.40W - Traffic: LOW - Ambient: 150 lux

Log Entry 11-15: Timestamp 2024-10-15T15:00:00 - Light Node SL_0015 - Status: DIMMED - Power: 11.50W - Traffic: LOW - Ambient: 150 lux

Log Entry 11-16: Timestamp 2024-10-15T16:00:00 - Light Node SL_0016 - Status: DIMMED - Power: 11.60W - Traffic: LOW - Ambient: 150 lux

Log Entry 11-17: Timestamp 2024-10-15T17:00:00 - Light Node SL_0017 - Status: DIMMED - Power: 11.70W - Traffic: LOW - Ambient: 150 lux

Log Entry 11-18: Timestamp 2024-10-15T18:00:00 - Light Node SL_0018 - Status: DIMMED - Power: 11.80W - Traffic: LOW - Ambient: 150 lux

Log Entry 11-19: Timestamp 2024-10-15T19:00:00 - Light Node SL_0019 - Status: DIMMED - Power: 11.90W - Traffic: LOW - Ambient: 150 lux

APPENDIX M: EXTENDED DATA LOGS

This section contains extended simulation logs and raw data outputs generated during the 24-hour simulation cycle. The data demonstrates the granular adjustments made by the system in response to micro-fluctuations in traffic and ambient light.

Log Entry 12-0: Timestamp 2024-10-15T00:00:00 - Light Node SL_0000 - Status: DIMMED - Power: 10.00W - Traffic: LOW - Ambient: 150 lux

Log Entry 12-1: Timestamp 2024-10-15T01:00:00 - Light Node SL_0001 - Status: DIMMED - Power: 10.10W - Traffic: LOW - Ambient: 150 lux

Log Entry 12-2: Timestamp 2024-10-15T02:00:00 - Light Node SL_0002 - Status: DIMMED - Power: 10.20W - Traffic: LOW - Ambient: 150 lux

Log Entry 12-3: Timestamp 2024-10-15T03:00:00 - Light Node SL_0003 - Status: DIMMED - Power: 10.30W - Traffic: LOW - Ambient: 150 lux

Log Entry 12-4: Timestamp 2024-10-15T04:00:00 - Light Node SL_0004 - Status: DIMMED - Power: 10.40W - Traffic: LOW - Ambient: 150 lux

Log Entry 12-5: Timestamp 2024-10-15T05:00:00 - Light Node SL_0005 - Status: DIMMED - Power: 10.50W - Traffic: LOW - Ambient: 150 lux

Log Entry 12-6: Timestamp 2024-10-15T06:00:00 - Light Node SL_0006 - Status: DIMMED - Power: 10.60W - Traffic: LOW - Ambient: 150 lux

COUNCIL FOR SKILLS AND COMPETENCIES
NATION BUILDING
THROUGH SKILLED YOUTH

Log Entry 12-7: Timestamp 2024-10-15T07:00:00 - Light Node SL_0007 - Status: DIMMED - Power: 10.70W - Traffic: LOW - Ambient: 150 lux

Log Entry 12-8: Timestamp 2024-10-15T08:00:00 - Light Node SL_0008 - Status: DIMMED - Power: 10.80W - Traffic: LOW - Ambient: 150 lux

Log Entry 12-9: Timestamp 2024-10-15T09:00:00 - Light Node SL_0009 - Status: DIMMED - Power: 10.90W - Traffic: LOW - Ambient: 150 lux

Log Entry 12-10: Timestamp 2024-10-15T10:00:00 - Light Node SL_0010 - Status: DIMMED - Power: 11.00W - Traffic: LOW - Ambient: 150 lux

Log Entry 12-11: Timestamp 2024-10-15T11:00:00 - Light Node SL_0011 - Status: DIMMED - Power: 11.10W - Traffic: LOW - Ambient: 150 lux

Log Entry 12-12: Timestamp 2024-10-15T12:00:00 - Light Node SL_0012 - Status: DIMMED - Power: 11.20W - Traffic: LOW - Ambient: 150 lux

Log Entry 12-13: Timestamp 2024-10-15T13:00:00 - Light Node SL_0013 - Status: DIMMED - Power: 11.30W - Traffic: LOW - Ambient: 150 lux

Log Entry 12-14: Timestamp 2024-10-15T14:00:00 - Light Node SL_0014 - Status: DIMMED - Power: 11.40W - Traffic: LOW - Ambient: 150 lux

Log Entry 12-15: Timestamp 2024-10-15T15:00:00 - Light Node SL_0015 - Status: DIMMED - Power: 11.50W - Traffic: LOW - Ambient: 150 lux

Log Entry 12-16: Timestamp 2024-10-15T16:00:00 - Light Node SL_0016 - Status: DIMMED - Power: 11.60W - Traffic: LOW - Ambient: 150 lux

Log Entry 12-17: Timestamp 2024-10-15T17:00:00 - Light Node SL_0017 - Status: DIMMED - Power: 11.70W - Traffic: LOW - Ambient: 150 lux

Log Entry 12-18: Timestamp 2024-10-15T18:00:00 - Light Node SL_0018 - Status: DIMMED - Power: 11.80W - Traffic: LOW - Ambient: 150 lux

Log Entry 12-19: Timestamp 2024-10-15T19:00:00 - Light Node SL_0019 - Status: DIMMED - Power: 11.90W - Traffic: LOW - Ambient: 150 lux

COUNCIL FOR SKILLS AND COMPETENCIES
NATION BUILDING
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APPENDIX N: EXTENDED DATA LOGS

This section contains extended simulation logs and raw data outputs generated during the 24-hour simulation cycle. The data demonstrates the granular adjustments made by the system in response to micro-fluctuations in traffic and ambient light.

Log Entry 13-0: Timestamp 2024-10-15T00:00:00 - Light Node SL_0000 - Status: DIMMED - Power: 10.00W - Traffic: LOW - Ambient: 150 lux

Log Entry 13-1: Timestamp 2024-10-15T01:00:00 - Light Node SL_0001 - Status: DIMMED - Power: 10.10W - Traffic: LOW - Ambient: 150 lux

Log Entry 13-2: Timestamp 2024-10-15T02:00:00 - Light Node SL_0002 - Status: DIMMED - Power: 10.20W - Traffic: LOW - Ambient: 150 lux

Log Entry 13-3: Timestamp 2024-10-15T03:00:00 - Light Node SL_0003 - Status: DIMMED - Power: 10.30W - Traffic: LOW - Ambient: 150 lux

Log Entry 13-4: Timestamp 2024-10-15T04:00:00 - Light Node SL_0004 - Status: DIMMED - Power: 10.40W - Traffic: LOW - Ambient: 150 lux

Log Entry 13-5: Timestamp 2024-10-15T05:00:00 - Light Node SL_0005 - Status: DIMMED - Power: 10.50W - Traffic: LOW - Ambient: 150 lux

Log Entry 13-6: Timestamp 2024-10-15T06:00:00 - Light Node SL_0006 - Status: DIMMED - Power: 10.60W - Traffic: LOW - Ambient: 150 lux

Log Entry 13-7: Timestamp 2024-10-15T07:00:00 - Light Node SL_0007 - Status: DIMMED - Power: 10.70W - Traffic: LOW - Ambient: 150 lux

Log Entry 13-8: Timestamp 2024-10-15T08:00:00 - Light Node SL_0008 - Status: DIMMED - Power: 10.80W - Traffic: LOW - Ambient: 150 lux

Log Entry 13-9: Timestamp 2024-10-15T09:00:00 - Light Node SL_0009 - Status: DIMMED - Power: 10.90W - Traffic: LOW - Ambient: 150 lux

Log Entry 13-10: Timestamp 2024-10-15T10:00:00 - Light Node SL_0010 - Status: DIMMED - Power: 11.00W - Traffic: LOW - Ambient: 150 lux

Log Entry 13-11: Timestamp 2024-10-15T11:00:00 - Light Node SL_0011 - Status: DIMMED - Power: 11.10W - Traffic: LOW - Ambient: 150 lux

Log Entry 13-12: Timestamp 2024-10-15T12:00:00 - Light Node SL_0012 - Status: DIMMED - Power: 11.20W - Traffic: LOW - Ambient: 150 lux

Log Entry 13-13: Timestamp 2024-10-15T13:00:00 - Light Node SL_0013 - Status: DIMMED - Power: 11.30W - Traffic: LOW - Ambient: 150 lux

COUNCIL FOR SKILLS AND COMPETENCIES
NATION BUILDING
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Log Entry 13-14: Timestamp 2024-10-15T14:00:00 - Light Node SL_0014 - Status: DIMMED - Power: 11.40W - Traffic: LOW - Ambient: 150 lux

Log Entry 13-15: Timestamp 2024-10-15T15:00:00 - Light Node SL_0015 - Status: DIMMED - Power: 11.50W - Traffic: LOW - Ambient: 150 lux

Log Entry 13-16: Timestamp 2024-10-15T16:00:00 - Light Node SL_0016 - Status: DIMMED - Power: 11.60W - Traffic: LOW - Ambient: 150 lux

Log Entry 13-17: Timestamp 2024-10-15T17:00:00 - Light Node SL_0017 - Status: DIMMED - Power: 11.70W - Traffic: LOW - Ambient: 150 lux

Log Entry 13-18: Timestamp 2024-10-15T18:00:00 - Light Node SL_0018 - Status: DIMMED - Power: 11.80W - Traffic: LOW - Ambient: 150 lux

Log Entry 13-19: Timestamp 2024-10-15T19:00:00 - Light Node SL_0019 - Status: DIMMED - Power: 11.90W - Traffic: LOW - Ambient: 150 lux

APPENDIX O: EXTENDED DATA LOGS

This section contains extended simulation logs and raw data outputs generated during the 24-hour simulation cycle. The data demonstrates the granular adjustments made by the system in response to micro-fluctuations in traffic and ambient light.

Log Entry 14-0: Timestamp 2024-10-15T00:00:00 - Light Node SL_0000 - Status: DIMMED - Power: 10.00W - Traffic: LOW - Ambient: 150 lux

Log Entry 14-1: Timestamp 2024-10-15T01:00:00 - Light Node SL_0001 - Status: DIMMED - Power: 10.10W - Traffic: LOW - Ambient: 150 lux

Log Entry 14-2: Timestamp 2024-10-15T02:00:00 - Light Node SL_0002 - Status: DIMMED - Power: 10.20W - Traffic: LOW - Ambient: 150 lux

Log Entry 14-3: Timestamp 2024-10-15T03:00:00 - Light Node SL_0003 - Status: DIMMED - Power: 10.30W - Traffic: LOW - Ambient: 150 lux

Log Entry 14-4: Timestamp 2024-10-15T04:00:00 - Light Node SL_0004 - Status: DIMMED - Power: 10.40W - Traffic: LOW - Ambient: 150 lux

Log Entry 14-5: Timestamp 2024-10-15T05:00:00 - Light Node SL_0005 - Status: DIMMED - Power: 10.50W - Traffic: LOW - Ambient: 150 lux

Log Entry 14-6: Timestamp 2024-10-15T06:00:00 - Light Node SL_0006 - Status: DIMMED - Power: 10.60W - Traffic: LOW - Ambient: 150 lux

COUNCIL FOR SKILLS AND COMPETENCIES
NATION BUILDING
THROUGH SKILLED YOUTH

Log Entry 14-7: Timestamp 2024-10-15T07:00:00 - Light Node SL_0007 - Status: DIMMED - Power: 10.70W - Traffic: LOW - Ambient: 150 lux

Log Entry 14-8: Timestamp 2024-10-15T08:00:00 - Light Node SL_0008 - Status: DIMMED - Power: 10.80W - Traffic: LOW - Ambient: 150 lux

Log Entry 14-9: Timestamp 2024-10-15T09:00:00 - Light Node SL_0009 - Status: DIMMED - Power: 10.90W - Traffic: LOW - Ambient: 150 lux

Log Entry 14-10: Timestamp 2024-10-15T10:00:00 - Light Node SL_0010 - Status: DIMMED - Power: 11.00W - Traffic: LOW - Ambient: 150 lux

Log Entry 14-11: Timestamp 2024-10-15T11:00:00 - Light Node SL_0011 - Status: DIMMED - Power: 11.10W - Traffic: LOW - Ambient: 150 lux

Log Entry 14-12: Timestamp 2024-10-15T12:00:00 - Light Node SL_0012 - Status: DIMMED - Power: 11.20W - Traffic: LOW - Ambient: 150 lux

Log Entry 14-13: Timestamp 2024-10-15T13:00:00 - Light Node SL_0013 - Status: DIMMED - Power: 11.30W - Traffic: LOW - Ambient: 150 lux

Log Entry 14-14: Timestamp 2024-10-15T14:00:00 - Light Node SL_0014 - Status: DIMMED - Power: 11.40W - Traffic: LOW - Ambient: 150 lux

Log Entry 14-15: Timestamp 2024-10-15T15:00:00 - Light Node SL_0015 - Status: DIMMED - Power: 11.50W - Traffic: LOW - Ambient: 150 lux

Log Entry 14-16: Timestamp 2024-10-15T16:00:00 - Light Node SL_0016 - Status: DIMMED - Power: 11.60W - Traffic: LOW - Ambient: 150 lux

Log Entry 14-17: Timestamp 2024-10-15T17:00:00 - Light Node SL_0017 - Status: DIMMED - Power: 11.70W - Traffic: LOW - Ambient: 150 lux

Log Entry 14-18: Timestamp 2024-10-15T18:00:00 - Light Node SL_0018 - Status: DIMMED - Power: 11.80W - Traffic: LOW - Ambient: 150 lux

Log Entry 14-19: Timestamp 2024-10-15T19:00:00 - Light Node SL_0019 - Status: DIMMED - Power: 11.90W - Traffic: LOW - Ambient: 150 lux

COUNCIL FOR SKILLS AND COMPETENCIES
NATION BUILDING
THROUGH SKILLED YOUTH